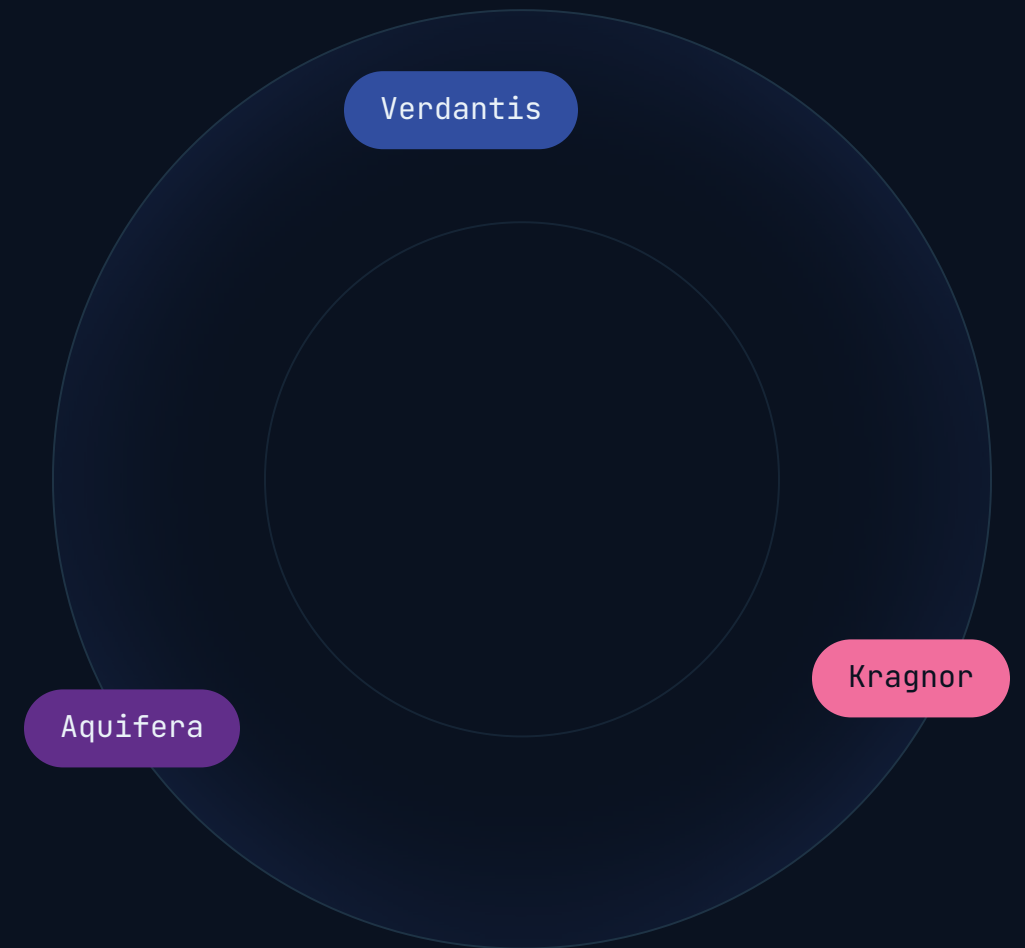


ocean Ai

// session 03 beginner Python

Build your galaxy

Store an ordered collection of worlds with a Python list.



Predict the branch.

If answer stores "n", what does `should_stop()` hand back?

True

False

"n"



You captain the ship. The engine still owns the galaxy.

your ship



engine planets



your choices

A list keeps items together in a fixed order.

Square brackets hold the collection. Commas separate each item.

0 · apple

1 · banana

2 · cherry

Three strings. One list.

Read the punctuation before reading the values.

```
fruits = ["apple", "banana", "cherry"]  
print(fruits)
```

.append() adds one item at the end.

Start empty, then watch the list grow in execution order.

```
planets = []  
planets.append("Verdantis")  
planets.append("Kragnor")  
planets.append("Aquifera")  
print(planets)
```



```
// predict count items
```

What will `len(planets)` print?

Verdantis

Kragnor

Aquifera

2

3

4


```
// reveal length
```

len() counts the items.

Positions start at zero; the count still starts at one.

```
planets = ["Verdantis", "Kragnor", "Aquifera"]  
print(len(planets))
```

OUTPUT 3

The first position is 0.

An index is the item's position inside square brackets.

0	1	2
Verdantis	Kragnor	Aquifera

One list can answer three lookups.

Trace each print against the position track.

```
planets = ["Verdantis", "Kragnor", "Aquifera"]  
print(planets[0])  
print(planets[1])  
print(planets[2])
```

OUTPUT Verdantis
Kragnor
Aquifera



```
// check valid positions
```

Three items means which indexes exist?

1, 2, 3

0, 1, 2

0, 1, 2, 3

Assign to an index to replace one item.

The list keeps its length and order; only position 1 changes.

```
planets = ["Verdantis", "Kragnor", "Aquifera"]  
planets[1] = "Nova Prime"  
print(planets)
```

in asks whether an item is present.

The answer is true or false, so last week's `if` can act on it.

```
planets = ["Verdantis", "Kragnor", "Aquifera"]

if "Kragnor" in planets:
    print("Kragnor is in your galaxy!")
else:
    print("No Kragnor here.")
```

The engine left two controls for us.

`False` keeps the engine galaxy. The empty list has no custom names yet.

```
USE_CUSTOM_PLANETS = False  
PLANETS = []
```

01 / enable

Switch to your galaxy.

Change only False to True.

```
USE_CUSTOM_PLANETS = True  
PLANETS = []
```


02 / build

Add five worlds in journey order.

The engine will attach fixed encounter details for now.

```
USE_CUSTOM_PLANETS = True
PLANETS = []
PLANETS.append("Verdantis")
PLANETS.append("Kragnor")
PLANETS.append("Aquifera")
PLANETS.append("Nova Prime")
PLANETS.append("Kato's Rest")
```


Build a galaxy worth exploring.

1 Enable custom planets.

2 Append at least five names.

3 Rename one item by index.

4 Run `uv run game.py`.

Success looks like

Your names appear in order. If the journey is longer, the engine cycles through them.

25 min

```
// verify inspect the collection
```

Check length, order, and membership.

Run these temporary checks, compare the output, then remove them.

```
print(len(PLANETS))  
print(PLANETS[0])  
print("Nova Prime" in PLANETS)
```

```
EXPECTED 5  
          Verdantis  
          True
```

IndexError means the position does not exist.

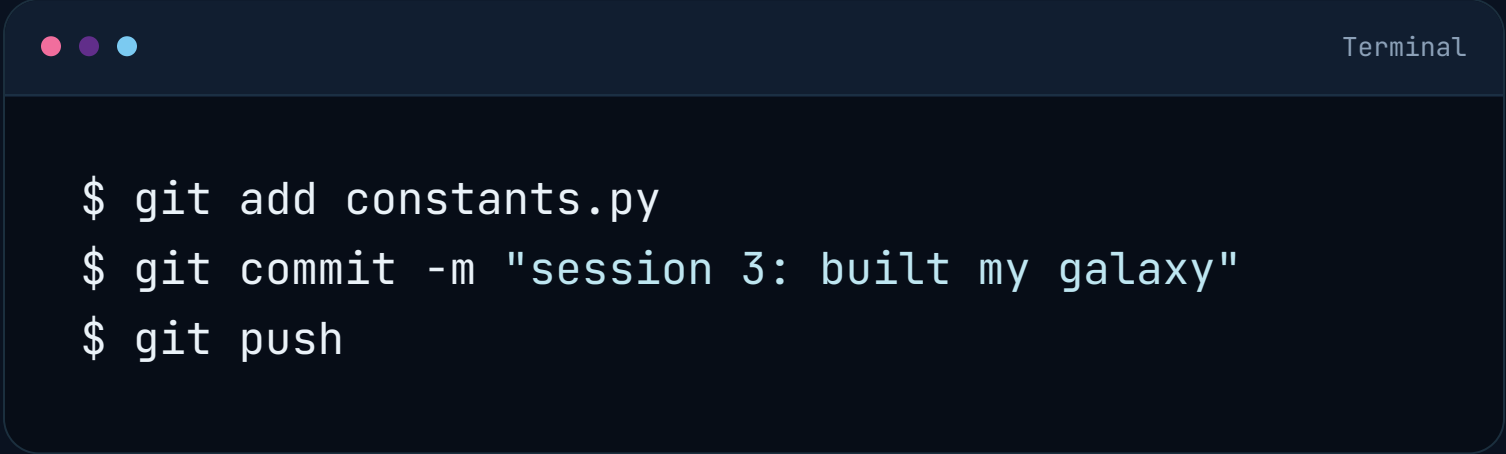
For five items, valid indexes are 0 through 4.

```
len(PLANETS) = 5  last index = 4
```

```
// save git checkpoint
```

Save your galaxy.

Run the journey once more and confirm the names first.

A terminal window with a dark blue background and a title bar with three colored dots (red, yellow, green) on the left and the word "Terminal" on the right. The terminal contains three lines of text:

```
$ git add constants.py  
$ git commit -m "session 3: built my galaxy"  
$ git push
```

```
$ git add constants.py  
$ git commit -m "session 3: built my galaxy"  
$ git push
```



```
// reflect ordered collections
```

What can a list do that one variable cannot?

- 01 How do you add an item?
- 02 Why does the first index equal zero?
- 03 Does GALAXY_SIZE still control your custom list?

Stretch the galaxy.

Choose one only after your commit.

EASY

Ten worlds

How does a longer journey feel?

MEDIUM

Trigger an error

Try index 99 and read the message.

PREVIEW

Print every world

Research a loop—we learn it next.

ocean Ai

// mission complete session 03

**You built the route.
Next, you fly it.**

Session 4 replaces the engine's travel loop with your code.

Session 04 • Loops